

P4: is U differentially algebraic? A negative result about the strategy, not about U

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Labels: PROVED means a complete argument, given the inputs it cites. VERIFIED means high-precision floating-point numerics, not an interval certificate. HEURISTIC means an argument without error control. Notation follows `paper/journal/paper2a.tex` (“paper2a”).

Summary of outcome

This note does *not* prove or disprove that U is differentially algebraic (D-algebraic), i.e. that it satisfies some $P(x, U, U', \dots, U^{(k)}) = 0$ for a polynomial $P \not\equiv 0$. It gives reasons to distrust the strategy proposed in `private/RESEARCH_PROGRAM_OPEN.md` row P4 — “dense poles on a positive-measure arc of $|x| = 1$ obstructs D-algebraicity, via a Nishioka/Malgrange-type theorem, anchored on Theorem `thm:fredholm`” — without exhibiting a decisive counterexample (§2: an attempted explicit counterexample is shown to be broken on inspection, post-review; what survives is the qualitative movable-singularities objection of Remark 1). It then identifies the correct class of theorems for this question (§3), checks their hypotheses against what `paper2a` actually proves (§5), and finds the single missing ingredient: a *quantitative* lower bound on the pole-counting function of U , which is not established anywhere in this programme. §4 explains why Theorem `thm:fredholm` is the wrong anchor: it is a linear (Fredholm/spectral) statement about a q -difference object, not a differential one, and the paper’s existing Nishioka-type results (`prop:noqdiff`, `cor:Qminusone`) already exhaust what difference-Galois methods say here — they say nothing about D -algebraicity. The row stays OPEN.

1 What is already proved, and what it does not give

Recall (`paper2a` §8, the “beyond transcendence” results):

- U, V are transcendental over $\mathbb{Q}(x)$ (`thm:U`, `thm:V`), not D -finite (`thm:nonDfinite`), satisfy no linear q -difference or Mahler equation (`prop:noqdiff`), and no $Q = -1$ relation (`cor:Qminusone`).
- $|x| = 1$ is a natural boundary for U, V (`thm:natboundary`), by Pólya–Bertrandias (`thm:PB`) applied to the fact that U, V are meromorphic on $|x| < 1$ with infinitely many poles there.
- The poles of U, V accumulate at *every* point of an explicit arc, $\arg x \in [\pi/6, 5\pi/6] \cup [7\pi/6, 11\pi/6]$ (`cor:arcUV`, from `thm:arc`). This is a statement about the closure of the pole set (topological density on the arc); it carries no information about how many poles lie in $|x| < r$ for a given $r < 1$, i.e. no counting-function estimate.

The mechanism behind `thm:nonDfinite` is Lemma `lem:odeposes`: a pole in $|x| < 1$ of a solution of a *linear* ODE with coefficients holomorphic on the closed disc must be a zero of the leading coefficient, hence there are only finitely many such poles unless the leading coefficient vanishes identically. Infinitely many poles accumulating on the boundary therefore rules out a linear ODE (with polynomial, or holomorphic-near- $\overline{D}$, coefficients). This is exactly why the row’s suggested next step reaches for the same kind of pole-count obstruction one dimension up, for *nonlinear* (algebraic) ODEs.

Remark 1 (Why this does not just generalize). Lemma `lem:odeposes` is special to linear equations: the singularities of a solution of a linear ODE are exactly the *fixed* singularities determined by the equation (zeros of the leading coefficient), independent of the solution or of initial data. Solutions of *nonlinear* algebraic ODEs can have *movable* singularities, whose location depends on the solution itself and need not be confined to a coefficient-determined finite set. Whether the movable singularities of a given algebraic ODE are poles (rather than branch points or worse) is exactly the content of Painlevé analysis / the Painlevé property, and equations that do have the Painlevé property (e.g. any order-1 equation $(y')^2 = \text{polynomial of degree } \leq 4 \text{ in } y$, the Weierstrass equation) have solutions whose movable poles can be *dense* on a curve. So “infinitely many poles dense on an arc” is not, by itself, evidence against D-algebraicity; the linear-case argument has no literal nonlinear analogue.

2 No verified counterexample; the real objection is movable singularities

Correction (post-review, Reviewer AH). An earlier version of this section claimed a PROVED explicit counterexample: a Weierstrass $\wp$ -function pulled back along $\zeta \mapsto x = e^{2\pi i \zeta / \omega_2}$ (ω_2 purely imaginary), asserting the sub-lattice $c + n\omega_1$ lands exactly on $|x| = 1$ with argument equidistributed by Weyl/Kronecker. This is arithmetically wrong: with ω_1 real and ω_2 purely imaginary, ω_1/ω_2 is purely *imaginary*, not real, so $\zeta/\omega_2 = c/\omega_2 + n(\omega_1/\omega_2) + m$ has its n -dependence entirely in the *modulus* of x (via $\Im(\zeta/\omega_2)$), not its argument; the sub-lattice does not stay on $|x| = 1$ at all, and shifting by the other generator changes x by a root of unity, not a dense set. Worse, the two requirements needed for the construction — $\tau = \omega_2/\omega_1$ non-real (so $\wp$ is a genuine, non-degenerate elliptic function) and ω_1/ω_2 real (so the pullback’s argument, not modulus, varies along a sub-lattice) — are mutually exclusive for any lattice Λ . Both attempts in the original write-up reduce, on inspection, to the ordinary q -expansion pullback $g(q) = \wp(\zeta)$, $q = e^{2\pi i \tau}$, whose poles accumulate only at $q = 0$ and $q = \infty$ (the cusps), never on a circle of positive radius — exactly the “first attempt” the original draft itself flagged as failing, restated with an incorrect modulus computation in the “fix”. **No explicit counterexample is exhibited here.** Proposition 2 below is downgraded from PROVED to a documented non-result, and Remark 1 — which does not depend on this construction — is the actual load-bearing objection to the ledger’s proposed strategy.

Proposition 2 (OPEN: no verified example, construction attempted and found broken). *We looked for a function g , D-algebraic (satisfying a fixed algebraic ODE with polynomial coefficients), meromorphic on a disc $|x| < 1$, with poles dense on a positive-measure arc of $|x| = 1$. We did not find one: the natural candidate (an elliptic function pulled back by an exponential change of variable along a lattice direction) provably cannot work, by the incompatibility noted above. Whether such a g exists (via, e.g., an automorphic/Fuchsian-group construction rather than a lattice pullback, or a genuinely different mechanism) is left open; we do not need it to make the point below, which rests instead on Remark 1.*

Consequently: *the ledger’s strategy is not vindicated by an exhibited counterexample*, but Remark 1 already gives an independent, qualitative reason to distrust it: Lemma `lem:odeposes`’s pole-confinement argument is a fact about *linear* ODEs (fixed singularities, at zeros of the leading coefficient, independent of the solution); solutions of *nonlinear* algebraic ODEs can have *movable* singularities whose location depends on initial data, and when an equation has the Painlevé property (e.g. the order-1 Weierstrass equation $(y')^2 = 4y^3 - g_2y - g_3$, or more generally any Painlevé-type equation) those movable singularities are honest poles, not merely branch points, with no a priori constraint tying their location to a coefficient-determined finite set. This is a structural reason the linear argument does not literally generalize; it is not by itself a proof that density-on-an-arc is compatible with D-algebraicity for some specific function, since we have not exhibited one. What it shows is that a theorem of the form “dense poles on a positive-measure boundary arc $\Rightarrow$ not D-algebraic” would have to say something genuinely new about movable singularities of nonlinear ODEs near a natural boundary – it is not a corollary of the elementary fixed-singularity fact that makes the linear case (`thm:nonDfinite`) work, and no such theorem is currently known to us in citable form. This matches the ledger’s own caveat that “some Nishioka theorems are about difference equations, not differential ones”: the density-only line of reasoning is not established to be sufficient, by any theorem we can cite, regardless of which named result one invokes.

3 The right class of theorem, and its hypotheses

The correct tool for turning growth/pole-density information about a meromorphic function into non-D-algebraicity is Nevanlinna-theoretic, not the fixed-singularity argument of `lem:odeposes`, and not Mahler/Nishioka difference-Galois theory (§4). The relevant literature is:

- Malmquist’s theorem (1913/1920) and its extensions: a first-order algebraic differential equation $F(x, y, y') = 0$ with a transcendental meromorphic solution on $\mathbb{C}$ of finite order forces the equation to be of a very restricted (Riccati-like) form.
- Steinmetz (*Ein Malmquistscher Satz für algebraische Differentialgleichungen höherer Ordnung*, J. Reine Angew. Math. 316 (1980)) and Bank–Kaufman (several papers, 1970s–80s, e.g. *On the growth of meromorphic solutions of nonlinear differential equations*, Trans. AMS 1980) generalize Malmquist’s theorem to higher order and give explicit bounds on the Nevanlinna characteristic $T(r, y)$ (equivalently, via the first main theorem, on the pole-counting function $N(r, y) = \int_0^r n(t, y)/t dt$) of any transcendental meromorphic solution of a fixed algebraic ODE with rational-function (or polynomial) coefficients, in terms of the degree of the coefficients and the order/degree of the equation.
- The upshot usable here is the contrapositive: if one can show that $n(r, U)$ (the number of poles of U in $|x| \leq r$) grows, as $r \rightarrow 1^-$, *faster* than any bound compatible with a solution of *some* algebraic ODE of bounded order and degree, for every order/degree pair, then U is not D-algebraic.

Remark 3 (the disc vs. the plane). The classical statements (Malmquist, Steinmetz, Bank–Kaufman) are for functions meromorphic on all of $\mathbb{C}$, where $T(r, y)$ is compared to $T(r)$ of the (polynomial or rational) coefficients as $r \rightarrow \infty$. U is meromorphic only on the disc $|x| < 1$, with $|x| = 1$ a natural boundary, so the relevant growth is of $n(r, U)$ or $T(r, U)$ as $r \rightarrow 1^-$ (necessarily $\rightarrow \infty$, since U is not rational). A full transplant of the Bank–Kaufman/Steinmetz machinery to functions meromorphic only on a disc, with polynomial ODE coefficients, is not something we could locate a ready-made citation for; the relevant framework is Tsuji’s Nevanlinna theory adapted to the disc (M. Tsuji,

Potential Theory in Modern Function Theory, 1959, Ch. III), and Bank himself has papers specifically on meromorphic solutions of algebraic differential equations *in the unit disc* (S. Bank, *On the instantaneous singularities of solutions of nonlinear differential equations*, and related 1970s work on the disc case), but we have not verified in detail that these give the precise counting-function bound needed here; this is exactly the “verify applicability carefully” caution the programme row asks for, and we flag it honestly as **HEURISTIC**: the shape of theorem is right, the citation for the disc case is not one we independently re-derived or checked line by line.

4 Why Theorem `thm:fredholm` does not anchor this

Theorem `thm:fredholm` computes $\det(I - \lambda\mathcal{T})$, a Fredholm determinant, in closed form as a q -cosine / Hahn–Exton q -Bessel function $F(\lambda, q) = \cos(\sqrt{\lambda}Z; q^2)$. Two things follow that are relevant here, and neither reaches D-algebraicity of $U(x)$:

1. It is a statement about F as a function of the spectral parameter λ at *fixed* q (an entire function of λ , an explicit q -Bessel function – D -finite in λ , in fact holonomic, since q -Bessel functions satisfy linear q -difference/differential equations in their argument). This says nothing about the dependence on q itself, which is where the natural boundary and the arc-density of `thm:arc` live.
2. As a function of q (equivalently x), $F(1, q) = 1 - \Sigma_1$ and its zeros are the travel poles; the operator $\mathcal{T} = \mathcal{T}(q)$ satisfies no linear q -difference equation in q that we are aware of and none is proved in paper2a – the “differential-Galois” structure available here is on the *difference* side (Nishioka’s theory, as cited at `prop:noqdiff` for Mahler equations, and the $Q = -1$ analysis of `cor:Qminusone`), not a differential-Galois structure in x . Difference-Galois/Nishioka theorems constrain algebraic relations and q -difference or Mahler equations satisfied by U ; paper2a already proves the strongest available statements of that kind (no linear q -difference or Mahler equation, no $Q = -1$ relation). None of this is evidence, for or against, an algebraic *differential* equation in x : a function can satisfy no Mahler/difference equation and still be, or fail to be, D-algebraic; the two properties are logically independent in general (Mahler functions and D-algebraic functions overlap only in degenerate cases, by the different normal forms Nishioka’s and Malmquist/Steinmetz’s theorems put on each).

So `thm:fredholm` is real machinery for the pole-existence and pole-density results already in the paper (via the explicit zero set of a q -Bessel function), but it is a statement in the wrong variable (the spectral parameter λ , or the difference-structure in q) to serve as the anchor for a differential-algebraicity argument in x . We could not find, and do not claim to have found, a route from `thm:fredholm` to a Nishioka-type differential-algebraic obstruction.

5 The precise gap, and what would close P4

Given §2–3, closing P4 (in either direction) needs:

- (a) *A quantitative rate.* A lower bound on $n(r, U)$, the number of poles of U in $|x| \leq r$, as $r \rightarrow 1^-$. `thm:arc`’s proof gives, near a fixed root of unity ζ on the arc, an infinite sequence of poles $q_j^* = \zeta e^{-t_j^*}$ with $t_j^* \sim t_j^0 = (L + i\pi)/(2j + 3/2)$ along an explicit tie ray, i.e. $q_j^* \rightarrow \zeta$ at rate $\sim c/j$. In principle this already gives, at each of the (dense set of) ζ on the arc where `cor:arcUV` applies, a sequence of poles approaching the boundary at a rate $O(1/j)$ in t , which

translates to $1 - |q_j^*| = O(1/j)$; summing over the countably many ζ with $N \geq 16$ has not been done in paper2a and does not obviously give a clean closed-form $n(r, U)$ estimate (poles from different ζ 's interleave in an uncontrolled way; no global counting argument is in the paper). This summation is the concrete, well-scoped computation that is missing; it looks tractable (each ζ contributes a $\sim \log(1/(1-r))$ -type count near itself, and there are $\sim r^{-1+\epsilon}$ -many admissible $\zeta = e(a/N)$, $N \lesssim 1/(1-r)$, on the arc) but was not carried out here, and doing it rigorously (as opposed to heuristically) would need to control double-counting and the N -dependence of the implied constants in `thm:arc`'s proof, which are not made explicit there (see `rem:arcUV`: “the index beyond which q_j^* is a pole ... is not made explicit”).

- (b) *The disc-analog counting theorem.* A Bank–Kaufman/Steinmetz-type upper bound, valid for meromorphic functions on a disc with a natural boundary, showing that no algebraic ODE of order k and degree d (for any fixed k, d) can have a solution meromorphic on $|x| < 1$ whose pole-counting function $n(r, \cdot)$ grows faster than [some explicit function of r, k, d] as $r \rightarrow 1^-$. We have identified the right family of theorems (§3) but have not verified that a version with the precise hypotheses needed (arbitrary k, d , natural boundary rather than entire coefficients) exists in the literature in citable form; `rem:disc` states this honestly as unresolved.
- (c) Compare (a) and (b) for a contradiction, for every k, d simultaneously (or via a uniform-in- (k, d) argument, e.g. showing $n(r, U)$ grows faster than any bound of the Bank–Kaufman shape, which typically depends polynomially or via a fixed exponential rate on k, d once $r \rightarrow 1$, whereas one would need superexponential growth in $1/(1-r)$ uniformly to beat all (k, d) at once, or a separate argument bounding the ODE's order/degree first, e.g. by degree considerations on residues as in `cor:Qminusone`'s use of 2-adic valuations).

None of (a), (b), (c) is done. **P4 is not closed by this note.** What this note contributes is negative-but-useful: it rules out the literal strategy stated in the ledger row (density on an arc $\Rightarrow$ non-D-algebraic, via `thm:fredholm`), replacing it with the correct (Nevanlinna/Bank–Kaufman/Steinmetz) family of theorems and pinpointing the exact missing input, a quantitative pole-counting rate together with a disc-version growth theorem, so that the next attempt does not repeat the invalid step of §2.

Weakest points

- §2's original elliptic-pullback construction is **wrong**, not merely under-polished: Reviewer AH caught that the claim “ $|x| = 1$ exactly, since ω_1 is real” (original line 132) is arithmetically false for the stated map, and that the underlying requirements ($\tau = \omega_2/\omega_1$ non-real for $\wp$ to be non-degenerate, vs. ω_1/ω_2 real for the pullback's argument to vary along $|x| = 1$) are mutually exclusive. This is fixed above by downgrading Proposition 2 to an open, unresolved question and relying instead on the qualitative Remark 1 (movable singularities / Painlevé property), which does not depend on the broken construction. A genuine explicit example (if one exists) would need a different mechanism, e.g. Fuchsian/automorphic rather than lattice-elliptic.
- §3/3: we cite Malmquist, Steinmetz, and Bank–Kaufman from memory of the theory's shape (Nevanlinna-theoretic growth bounds for meromorphic solutions of algebraic ODEs); we did not independently re-derive or numerically check any of these statements in this session, and we explicitly flag that the disc/natural-boundary version of the theory is the part we

could not pin down a precise citable statement for. Treat the bibliographic pointers here as HEURISTIC memory, to be checked against the primary sources before citing in a paper.

- §5(a)'s claimed shape of a global counting estimate ($n(r, U) \sim$ combining $\sim r^{-1}$ many arcs each contributing a logarithmic count) is a back-of-envelope estimate, not a proof; it is plausible from `thm:arc`'s local rate but we did not attempt the summation.
- No computation (Rust or otherwise) was run for this note: the question, as scoped, turned out to be about which theorem applies and why, not about a numerical question the existing tools (`tlseries`, `u5b`, etc.) could resolve. If a future pass pursues (a), a Rust enumeration of travel-pole counts $n(r, U)$ at a sequence of $r \rightarrow 1^-$, cross-checked against the explicit t_j^* formula, would be the natural first computational step, wrapped under the standing `perl -e 'alarm 290; exec @ARGV' / ≤ 3GB / fresh-sysctl` discipline.